KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · MATH

The Motor City Robot Rally

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Scoring the Robots

Kavya's robotics team is competing at a robot rally in Detroit, Michigan. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Unit 5): a robot's wheel guard is a regular hexagon with sides of 6 centimeters. a) What is its perimeter? b) How many lines of symmetry does it have?

2. Kavya's robot places 3 cones worth 4 points each. Write a multiplication sentence for the cone points, and find the total.

3. Kamau says 2 + 3 × 4 = 20. Kendrick says it equals 14. Who do you think is right? Explain your thinking.

4. What do you think the parentheses tell you to do first in (5 + 5) × 3? Then find the value.

5. Which expression matches "add 6 and 2, then multiply by 5"?

   (6 + 2) × 5 | 6 + 2 × 5 | 6 × 2 + 5

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Judge the Scores

Grouping symbols first, then multiply and divide from left to right, then add and subtract from left to right. Show every step.

1 Evaluate each expression. a) 40 − 6 × 5; b) 72 ÷ 8 × 3; c) 2 × [18 − (5 + 4)]

2 Write an expression for each, but do not evaluate it. a) subtract 8 from 50, then divide by 6; b) four times as large as 327 + 85

3 Two steps: a robot places 9 cones worth 3 points each and 5 cubes worth 4 points each, then loses 6 points for a penalty. Write one expression for the score, then evaluate it.

4 Find the mistake: Kendrick says 36 − 12 + 4 = 20 because you add before you subtract. In two or three complete sentences, explain his mistake and give the correct value.

★ **Challenge:** Add one pair of parentheses to make each equation true. a) 3 + 5 × 4 − 2 = 30; b) 48 ÷ 4 + 2 × 3 = 24

Looking Ahead: Week 27: Powers of 10 and Exponents. At the Adler Planetarium in Chicago, you will multiply and divide by powers of 10 and write 10 × 10 × 10 as 10^3.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · READING

Gold Above the Rink

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Stories Behind Our Words

Today you will skate at Rockefeller Center in New York City with Meera and her cousin, and read myths from ancient Greece, Rome, and Hawai'i. Show what you already know. It is fine to guess!

1. Review (Unit 5): Read: "Meera gripped the rail with both hands and would not let go." Make an inference about how Meera feels, and write the clue that tells you.

2. A **myth** is a traditional story, often about gods or heroes. Write the name of any myth, legend, or hero story you have heard of, and one thing that happens in it.

3. Maddox says, "Carrying everyone's skates up the subway stairs was a Herculean job." Hercules was a hero who did twelve almost impossible tasks. What does Maddox mean?

4. Which phrase do you think means "one weak spot in someone strong"?

   An Achilles' heel | A Midas touch | Pandora's box

5. Many myths explain something in nature. Write one question about nature (for example, about the sun, fire, or echoes) that a myth might try to answer.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · READING

Gold Above the Rink

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Allusion Detective

Write in complete sentences with correct capitals, commas, apostrophes, and quotation marks.

1. Read: "After Maddox spilled the hot cocoa, he complained about the cold, and then he lost his gloves. 'This trip has turned into Pandora's box,' he sighed." Explain what Maddox means AND tell the myth behind the allusion.

2. Two steps. Aunt Maren is a great skater, but she is afraid of skating backward. Which allusion fits her fear? Write a sentence that uses it correctly.

3. Find the mistake. Mehdi wrote: "Narcissus could only repeat the last words he heard, so Juno turned him into a flower." Rewrite the retelling so it is accurate.

4. Be the editor. Rewrite each sentence correctly. a) Later Midas washed in the river. b) Meera asked, "can we skate at rockefeller center again?"

★ **Challenge:** In 3 or 4 sentences, retell the Hawaiian moʻolelo of Māui and the sun in your own words. How is it like the myth of Prometheus?

Looking Ahead: Week 27: Mystery stories set in Savannah, Georgia. You will hunt for clues, spot red herrings (clues that point the wrong way), make inferences, and compare how two mysteries handle the same theme.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · SCIENCE

Gravity: Why Things Fall

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 24):** Name the three kinds of plate boundaries. Then explain what happens to an ocean plate at a subduction zone.

2. Why do you think a dropped ball falls down instead of up or sideways? Write your best idea.

3. Predict: On the Moon, would you weigh the same as on Earth? Would your body have the same amount of matter? Explain.

4. Draw Earth as a circle with three people standing in different places around it. Draw an arrow from each person's feet showing which way "down" is.

5. Predict: Does a falling ball move at the same speed the whole way down, or does its speed change? Why?

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · SCIENCE

Gravity: Why Things Fall

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Measuring the Pull

Use evidence from today's lesson. Write in complete sentences.

1. **Mass or weight?** A 3-kilogram melon: mass on the Moon? Weight on Earth (9.8) and on the Moon (1.6)?

2. **Read the data.** NASA lists Mars at 3.7 and Saturn (cloud level) at 9.0 newtons on each kilogram, though Saturn has about 95 times Earth's mass. Use mass AND distance to explain.

3. **Two-step.** With no air, how fast is a falling object moving after 2 seconds? How much faster after 5?

4. **Find the mistake.** Padma says, "A pan balance would show a smaller mass on the Moon." What is wrong?

★ **Challenge: Explain.** NASA says gravity at the space station is about 90 percent as strong as on the ground. So why do astronauts float?

Looking Ahead: Next week: Earth's Water by the Numbers. How much water is salty, fresh, frozen and underground? You will graph it like a scientist.

KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · SOCIAL STUDIES

The Civil Rights Movement (1950s–1960s)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Unit 5):** The Double V campaign of 1942 asked for two victories. What were they, and which one was still unfinished when World War II ended?

2. What do you already know about the civil rights movement? Write two facts or guesses.

3. Prediction: A boycott means refusing to buy or use something. How could staying off city buses change a city's unfair law?

4. Why might a child or a teenager be able to change history? Explain your thinking.

5. Write one question you have about the civil rights movement.

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The Civil Rights Movement (1950s–1960s)

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Courage That Changed the Law

Use evidence from today's lesson. Write in complete sentences.

1. **Put in order with years:** Voting Rights Act; Mendez v. Westminster; Montgomery bus boycott begins; Brown v. Board of Education; Little Rock Nine enter Central High.

2. **Find the mistake.** "Rosa Parks was the first person arrested for refusing to give up her seat on a Montgomery bus." Rewrite the sentence so it is true.

3. **Compare two sources.** The Supreme Court (1954) said separate schools are "inherently unequal." Eisenhower (1957) said "Mob rule cannot be allowed to override the decisions of our courts." How are they connected?

4. **Write a paragraph.** Choose Sylvia Mendez, Claudette Colvin, Ruby Bridges, Harry T. Moore or John Lewis. In 3 or 4 sentences, tell the risk this person took and what changed. Check capitals, commas and end punctuation.

★ **Challenge: Two steps:** Harry T. Moore started a branch of the NAACP in 1934. The Civil Rights Act came 30 years later. What year was that, and what is one thing it outlawed?

Looking Ahead: Next week: more voices for equal rights, including the Delano grape strike, Indigenous self-determination, Title IX and the 1977 Section 504 sit-in.